Supplementary Methods and Reproducibility Attachments

Descriptive post-amendment analysis; complete original confirmation requirements were not met.

Detailed methods

The full original frozen protocol and main Methods define all mathematical objectives, feature groups, solver/numerical policies, inference and sensitivities. This supplement supplies complete animal scores and all registered panel/reference summaries; machine-readable tables retain every computed cell and fullprecision.

A1 Protocol and amendments

Original protocol hash and preserved source freezes; pre-D scope change after E/generator/coverage evidence; all broad confirmation withheld. OWNER_MISSION and PRE_D_STOPPING_RULE_AMENDMENT document prospective authority.

A2 Dataset manifest

DANDI000473/0.230417.1502; original25animals,19fixed candidates/10development9final; original asset/ETag/epoch/unit and all exclusions in old EPOCH_UNIT_ALLOCATION and QC journals; no source/animal replacement.

A3 Pipeline and environment

Raw timestamp reader -> C-only selected QC/counts -> frozen Cscalers/models -> one D/count/score bundle -> fixed mouse statistics/controls/sensitivities -> saved figures. Original scientific models/solver/AD rules preserved; completeCmanifest contains198fit identities/594certificates.

A4 All baselines and animals

All B0-B4,D,C,S,Bstar scores are in the full tables below and animal_model_scores.csv; no favorable baseline or mouse pruning. Bstar=B2 and sharedoffset=B4 are distinct.

A5 Holdout audit

C-only admission/scaling/fitting precedes D; Dstates/once marker/exportattempts/sourceandactionjournals preserve one assessment. Trusted raw fullrecord decode is disclosed; no D outcome selects models. No second D or repaired evaluation is authorized.

A6 Statistics

Exact inclusive sign tests require conditional fair signs, not arbitrary meanzero. Holm3primary and Holm3guards,10000jointwholemousebootstrap/linear95CI,directionandLOMO all retained; fullprecision CSV accompanies rounded display. Nonsignificance is not equivalence.

A7 Falsification and failed reference

Three original999stress suites reported for E/D; conditionalBstar failed at156130rep0and remains unavailable. Diagnostic prefix excluded from originalstatistics; all historicalpartialarrays/failure receipts preserved. No replacementnull or clippedsimulation.

A8 QC/leakage/sensitivity

Original C-onlyphysiological screen; exactunit/trialmasks/scalers; fixedparentalignedwindows exclude responsebins; allmandatory sensitivities and failed6of9recent coverage retained. Singleprobe/singlelab/unknownstate andsorting/anatomylimits remain.

A9 Reproduction

Saved-only reproduction verifies source/result hashes and renders figures without source access/fits/D rescoring. Original complete scientific entrypoints are consumed; full-executable path is documented but not rerun. Author uses current final saved evidence only.

A10 Claim-to-evidence

Held-out descriptive gain supported only by saved9mouse Dscores; conditional statistical evidence by exactregistered summaries under assumptions; Bstar/S comparisons retain guards; originalM6conditional reference failed andcoverageinsufficient. All broadconfirmation/validatedtemporal/NAP/CWP/causal/independentlabclaims withheld.

Full E and D animal model scores

Partition	Mouse	B0	B1	B2	B3	B4	D	C	S	Bstar
E	147465	-0.67370461	-0.67390283	-0.63854048	-0.65675855	-0.64596496	-0.63035599	-0.64409229	-0.63099215	-0.63854048
E	152417	-0.82984858	-0.82343251	-0.78078939	-0.83030434	-0.80636492	-0.79772615	-0.80465788	-0.79495805	-0.78078939
E	152419	-0.96483723	-0.94884758	-0.86039502	-0.90015738	-0.84027155	-0.81612178	-0.84176545	-0.80754925	-0.86039502
E	156130	-0.46599582	-0.46671243	-0.41151536	-0.46456547	-0.46030287	-0.45879434	-0.45984559	-0.46031011	-0.41151536
E	156131	-0.88435376	-0.88105236	-0.81427394	-0.87505341	-0.81929904	-0.80751021	-0.8170225	-0.80971674	-0.81427394
E	216300	-0.62032883	-0.62273809	-0.59081505	-0.61461451	-0.61888124	-0.61583286	-0.61807693	-0.61887809	-0.59081505
E	216301	-0.68344411	-0.68222963	-0.65463304	-0.67319516	-0.6536674	-0.65281213	-0.65231153	-0.65366628	-0.65463304
E	258416	-1.0621019	-1.0723682	-1.0350182	-1.0649865	-1.1219219	-1.1114561	-1.1195937	-1.1216127	-1.0350182
E	268951	-0.90923388	-0.93087333	-1.2280369	-0.88681813	-0.97882595	-0.92472596	-0.97981605	-0.9421025	-1.2280369
E	273858	-0.55348799	-0.53354254	-0.51126372	-0.56260893	-0.52833732	-0.52323319	-0.52975145	-0.52757422	-0.51126372
D	128514	-0.99374314	-0.99145131	-0.92566229	-0.98428503	-0.93133476	-0.89304974	-0.93017402	-0.90248977	-0.92566229

Partition	Mouse	B0	B1	B2	B3	B4	D	C	S	Bstar
D	128516	-0.63430366	-0.61849683	-0.56092728	-0.61298475	-0.59769449	-0.57533421	-0.60417384	-0.59252183	-0.56092728
D	152414	-1.0154719	-1.0137259	-0.98971957	-1.0021947	-0.97747725	-0.95342954	-0.97967472	-0.95567053	-0.98971957
D	225757	-0.59826767	-0.59671422	-0.55878765	-0.59429046	-0.56035469	-0.55391981	-0.55861851	-0.55691464	-0.55878765
D	225758	-1.1794724	-1.1451407	-1.0485016	-1.1168233	-1.0113453	-0.99659612	-1.0234072	-0.99544257	-1.0485016
D	225759	-0.87776989	-0.88164085	-0.77283531	-0.8147964	-0.78804166	-0.74625787	-0.78416228	-0.75283321	-0.77283531
D	258414	-1.0239443	-1.0106458	-0.9555345	-1.0055318	-0.97134318	-0.96600146	-0.97317811	-0.96502857	-0.9555345
D	268947	-0.63305809	-0.63437284	-2.2341041	-0.5987812	-0.54472307	-0.53530498	-0.54545119	-0.54198895	-2.2341041
D	273855	-0.60252384	-0.5955985	-0.52441604	-0.59651598	-0.5074734	-0.50501963	-0.50568205	-0.5074727	-0.52441604

All six contrast statistics

Partition	Contrast	Mean	Raw p	Holm p	CI low	CI high	Positive/n	Min LOMO
E (exploratory)	D_minus_B4	0.013526846	0.001953125	0.005859375	0.005870354	0.024250325	10/10	0.0090187184
D (descriptive)	D_minus_B4	0.01831938	0.00390625	0.01171875	0.0099289438	0.027586852	9/9	0.015386329
E (exploratory)	C_minus_B4	0.00069037438	0.17578125	0.17578125	-0.00020635951	0.0015380056	7/10	0.00050839499
D (descriptive)	C_minus_B4	-0.0016371285	0.390625	0.390625	-0.0049570008	0.0010831615	4/9	-0.0023266921
E (exploratory)	D_minus_C	0.012836471	0.00390625	0.0078125	0.0049188008	0.024010343	9/10	0.0081416243
D (descriptive)	D_minus_C	0.019956509	0.00390625	0.01171875	0.011041037	0.028715223	9/9	0.017713021
E (exploratory)	D_minus_Bstar	0.01867124	0.85351562	1	-0.028664065	0.090052805	5/10	-0.012955392
D (descriptive)	D_minus_Bstar	0.20506389	0.03515625	0.0703125	0.0080356347	0.58298071	7/9	0.01834698
E (exploratory)	C_minus_Bstar	0.0058347687	0.97851562	1	-0.035522547	0.066725865	3/10	-0.021097017
D (descriptive)	C_minus_Bstar	0.18510738	0.63671875	0.63671875	-0.012751876	0.56487991	5/9	-0.0028358159
E (exploratory)	D_minus_S	0.0028791455	0.23632812	0.70898438	-0.0010289439	0.0072031611	8/10	0.0012683238
D (descriptive)	D_minus_S	0.0050499334	0.015625	0.046875	0.001772565	0.0089124157	7/9	0.0035327226

Full retained reference summaries

Part	Suite	Contrast	Mean ref	q95 ref	Observed	Obs-ref
E	history	D_minus_B4	0.011303113	0.011987263	0.013526846	0.0022237325

Part	Suite	Contrast	Mean ref	q95 ref	Observed	Obs-ref
E	history	C_minus_B4	-0.0035711296	-0.0031752362	0.00069037438	0.0042615039
E	history	D_minus_C	0.014874243	0.015677375	0.012836471	-0.0020377715
E	history	D_minus_Bstar	0.03815204	0.040358497	0.01867124	-0.0194808
E	history	C_minus_Bstar	0.023277798	0.025484044	0.0058347687	-0.017443029
E	history	D_minus_S	-0.0039121399	-0.0031802054	0.0028791455	0.0067912854
E	task_order	D_minus_B4	0.013443923	0.013532199	0.013526846	8.2922798e-05
E	task_order	C_minus_B4	0.00073520207	0.00083302626	0.00069037438	-4.4827687e-05
E	task_order	D_minus_C	0.012708721	0.012892804	0.012836471	0.00012775049
E	task_order	D_minus_Bstar	0.018844605	0.019622073	0.01867124	-0.00017336449
E	task_order	C_minus_Bstar	0.0061358837	0.0069012771	0.0058347687	-0.00030111498
E	task_order	D_minus_S	0.0028055707	0.002876703	0.0028791455	7.3574879e-05
E	temporal	D_minus_B4	0.011653552	0.011875295	0.013526846	0.0018732934
E	temporal	C_minus_B4	0.00063839188	0.00077288305	0.00069037438	5.1982495e-05
E	temporal	D_minus_C	0.011015161	0.011275031	0.012836471	0.0018213109
E	temporal	D_minus_Bstar	0.011267386	0.067855436	0.01867124	0.0074038541
E	temporal	C_minus_Bstar	0.00025222549	0.056950994	0.0058347687	0.0055825432
E	temporal	D_minus_S	0.0031139844	0.0033819034	0.0028791455	-0.00023483882
D	history	D_minus_B4	0.013770829	0.014414355	0.01831938	0.0045485512
D	history	C_minus_B4	-0.0085298972	-0.008090522	-0.0016371285	0.0068927687
D	history	D_minus_C	0.022300726	0.023075748	0.019956509	-0.0023442175
D	history	D_minus_Bstar	0.23364494	0.23561907	0.20506389	-0.028581051
D	history	C_minus_Bstar	0.21134421	0.21335032	0.18510738	-0.026236833
D	history	D_minus_S	-0.0084161226	-0.0075546655	0.0050499334	0.013466056
D	task_order	D_minus_B4	0.018291376	0.018488863	0.01831938	2.8004412e-05
D	task_order	C_minus_B4	-0.0016229053	-0.001214706	-0.0016371285	-1.4223233e-05
D	task_order	D_minus_C	0.019914281	0.020430962	0.019956509	4.2227645e-05
D	task_order	D_minus_Bstar	0.2040243	0.20481231	0.20506389	0.0010395852
D	task_order	C_minus_Bstar	0.18411002	0.18517276	0.18510738	0.00099735757

Part	Suite	Contrast	Mean ref	q95 ref	Observed	Obs-ref
D	task_order	D_minus_S	0.0049929991	0.005129288	0.0050499334	5.6934296e-05
D	temporal	D_minus_B4	0.015573464	0.015825176	0.01831938	0.0027459161
D	temporal	C_minus_B4	-0.0011330376	-0.00086825562	-0.0016371285	-0.00050409089
D	temporal	D_minus_C	0.016706502	0.0171445	0.019956509	0.0032500069
D	temporal	D_minus_Bstar	0.12365464	0.27027715	0.20506389	0.081409246
D	temporal	C_minus_Bstar	0.10694814	0.25317144	0.18510738	0.078159239
D	temporal	D_minus_S	0.0036979212	0.0040080379	0.0050499334	0.0013520122

All sensitivity panels

Panel	Mandatory	Qualified	Computed	Coverage	D-B4	C-B4	D-C	D-Bstar	C-Bstar	D-S
alignment_minus01	True	9	9	True	0.018617646	-0.0014282839	0.02004593	0.1909282	0.17088227	0.0051708613
alignment_plus01	True	9	9	True	0.018311026	-0.0017535718	0.020064598	0.21299255	0.19292796	0.0049374154
deep_PL5_PL6a	True	9	9	True	0.01833942	-0.0014630543	0.019802474	0.20004818	0.1802457	0.0050697528
first4	True	9	9	True	0.024370224	-0.0017556385	0.026125863	0.013630219	-0.012495643	0.010263129
higher_raw_DV_rank4	False	9	9	None	0.013283219	0.00037824499	0.012904974	0.38633384	0.37342887	0.0040808513
label_PL	False	0	0	None	UNAVAILABLE	UNAVAILABLE	UNAVAILABLE	UNAVAILABLE	UNAVAILABLE	UNAVAILABLE
label_PL1	False	0	0	None	UNAVAILABLE	UNAVAILABLE	UNAVAILABLE	UNAVAILABLE	UNAVAILABLE	UNAVAILABLE
label_PL2/3	False	0	0	None	UNAVAILABLE	UNAVAILABLE	UNAVAILABLE	UNAVAILABLE	UNAVAILABLE	UNAVAILABLE
label_PL4	False	0	0	None	UNAVAILABLE	UNAVAILABLE	UNAVAILABLE	UNAVAILABLE	UNAVAILABLE	UNAVAILABLE
label_PL5	False	2	2	None	0.031166362	-0.00051836965	0.031684731	0.034451289	0.0027665575	0.0058405074
label_PL6a	False	7	7	None	0.014674579	-0.0017329642	0.016407543	0.24736158	0.23095403	0.0048495372
label_PL6b	False	0	1	None	UNAVAILABLE	UNAVAILABLE	UNAVAILABLE	UNAVAILABLE	UNAVAILABLE	UNAVAILABLE
lower_raw_DV_rank4	False	9	9	None	0.024370224	-0.0017556385	0.026125863	0.013630219	-0.012495643	0.010263129
provider_active	False	9	9	None	0.018306024	-0.0020537527	0.020359777	0.20783134	0.18747157	0.0051897016
quality005	True	9	9	True	0.01831938	-0.0016371285	0.019956509	0.20506389	0.18510738	0.0050499334
rate4	True	9	9	True	0.010962464	0.0013424786	0.0096199851	0.0075211007	-0.0020988844	0.0043850724
recent_event	True	6	7	False	0.016001891	-0.0021164246	0.018118316	0.0083204076	-0.0097979085	0.0062962049
session_order_quartile1	False	0	0	None	UNAVAILABLE	UNAVAILABLE	UNAVAILABLE	UNAVAILABLE	UNAVAILABLE	UNAVAILABLE

Panel	Mandatory	Qualified	Computed	Coverage	D-B4	C-B4	D-C	D-Bstar	C-Bstar	D-S
session_order_quartile2	False	0	0	None	UNAVAILABLE	UNAVAILABLE	UNAVAILABLE	UNAVAILABLE	UNAVAILABLE	UNAVAILABLE
session_order_quartile3	False	9	9	None	0.016516547	0.00033891195	0.016177635	0.03571132	0.019533685	0.0031954965
session_order_quartile4	False	9	9	None	0.020122214	-0.003613169	0.023735383	0.37441645	0.35068107	0.0069043703

Anatomical and dependence summaries

Variable	Group	n	D-B4	C-B4	D-C
condition	Aversion	9	0.01831938	-0.0016371285	0.019956509
genotype	NOT_DECODED_FROM_SOURCE; field present in schema, value not examined	9	0.01831938	-0.0016371285	0.019956509
individual_probe	128514 / probe0	1	0.038285012	0.0011607337	0.037124278
individual_probe	128516 / probe0	1	0.022360276	-0.006479353	0.028839629
individual_probe	152414 / probe0	1	0.024047711	-0.002197473	0.026245184
individual_probe	225757 / probe0	1	0.0064348764	0.0017361738	0.0046987026
individual_probe	225758 / probe0	1	0.014749167	-0.01206191	0.026811077
individual_probe	225759 / probe0	1	0.04178379	0.0038793803	0.03790441
individual_probe	258414 / probe0	1	0.0053417247	-0.0018349304	0.0071766551
individual_probe	268947 / probe0	1	0.0094180864	-0.00072812842	0.010146215
individual_probe	273855 / probe0	1	0.0024537784	0.0017913507	0.00066242773
laboratory	SOURCE_STUDY_SINGLE_LABORATORY	9	0.01831938	-0.0016371285	0.019956509
raw_provider_DV_median	-2.09932127452042	1	0.0094180864	-0.00072812842	0.010146215
raw_provider_DV_median	-2.3860768750885	1	0.024047711	-0.002197473	0.026245184
raw_provider_DV_median	-2.40153620662112	1	0.0024537784	0.0017913507	0.00066242773
raw_provider_DV_median	-2.4718832356413802	1	0.0064348764	0.0017361738	0.0046987026
raw_provider_DV_median	-2.47433937267515	1	0.0053417247	-0.0018349304	0.0071766551
raw_provider_DV_median	-2.53821414185162	1	0.04178379	0.0038793803	0.03790441
raw_provider_DV_median	-2.542905164118025	1	0.038285012	0.0011607337	0.037124278
raw_provider_DV_median	-2.543293750672285	1	0.022360276	-0.006479353	0.028839629
raw_provider_DV_median	-2.6388757992260903	1	0.014749167	-0.01206191	0.026811077

Full six-contrast group and leave-group-out means, including empty remainders, are retained in descriptive_groups.csv and original FINAL_D_RESULTS.json. Raw coordinate groups do not establish physical-depth equivalence; missing genotype means not decoded, not provider absence. These are not new hypothesis tests.

Generator failure and numerical history

Original conditional B2 exponential failed on a reproduced feedback path; all old P12M4andP13failures remain unchanged. No mathematical repair, count censoring or different null was used. Current C198fits are a separate finite observed-data numerical result, not evidence that recursive B2simulation is stable.

Author review requirements

Confirm author/affiliation/contribution/conflict and ethics/data-use statements; audit scientific wording, tables and declared assumptions; approve any submission separately. Internal agent review is not external peer review.

Reproduction boundary

Run the saved-only PowerShell reproduction supplied with this task. It does not refit C, export raw D, rescore D, generate a new real control, select models or consume a second assessment. Machine-readable CSV/JSON/manifests and preserved old archives contain all per-cell details; no hypothetical model recovery is executed.

Exact saved-only reproduction and post-execution audit note

From the repository root in Windows PowerShell, run: & .\paper13_descriptive_final1\reproduce_closeout.ps1

The script uses the verified scientific Python environment, preserved original derived A/C/B/E states and Autograd1.8.0 dependency. An explicit Python path can be supplied with -Python. See paper13_descriptive_final1/REPRODUCTION.md for environment paths, hash-verified saved-state restoration and separate-directory table/figure/PDF regeneration. Raw-source acquisition, real C/D entrypoints and new real control draws are not reproduction steps; their scientific entitlement is consumed.

The first saved-only post-D audit passed88synthetic tests but then used an incorrect bitwise oracle for differently associated floating sums. The original score computed a single unit-level total, while saved joint-bin files had already summed eight units;22of81mean comparisons differed by at most2ULP(2.22e-16). All18same-order D/S means matched exactly. The failed receipt and all638pre-D frozen files remain unchanged. A separately registered saved-only audit correction and eight added regressions bring the current total to96unique synthetic tests. No model, score, scientific tolerance, statistic or real assessment was altered.

The corrected audit verifies original per-mouse/final/journal score copies, hashes, B*/B2aliases and all six statistics exactly, retaining exact D/S reductions. Other aggregates receive a fixed exact-rational float64 forward-error compatibility check. It explicitly assumes unseen per-unit computed log terms are nonpositive: joint-bin files alone cannot establish that premise or recover individual unit floats losslessly. This is an archival integrity/conditional aggregate check, not a new independent likelihood evaluation, scientific numerical policy or retrospective PASS of the failed null. The registration, derivation, failed attempt, regressions and independent

review remain in the mission evidence directory.